

Regulator Narrative

Purpose

This document provides a regulator-facing narrative explaining how the AI Trust & Risk platform operates in a controlled, auditable, and human-governed manner. It is written to support regulatory review, licensing, audits, and supervisory discussions.

The system is designed to **assist human decision-making**, not replace it, and to operate within strict governance, transparency, and accountability constraints.

1. Regulatory Design Principles

The platform is built on five core regulatory principles:

1. Human Accountability First

- 2. AI never takes autonomous enforcement actions
- 3. Humans remain the final decision-makers

4. Explainability by Design

- 5. Every AI output is traceable, reproducible, and explainable

6. Deterministic & Predictable Behavior

- 7. No black-box machine learning models in production
- 8. Rule-based, versioned, and auditable logic

9. Continuous Oversight & Monitoring

- 10. Ongoing drift detection, accuracy tracking, and health scoring

11. Controlled Change Management

- 12. No AI behavior changes without explicit human approval
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2. System Scope & Boundaries

The AI system operates strictly as a **decision-support layer** within Trust Operations.

Explicit Non-Capabilities

-  No automated blocking of users
-  No automated account suspension
-  No autonomous financial decisions
-  No self-learning or self-modifying behavior

Permitted Capabilities

-  Risk assessment
 -  Behavioral analysis
 -  Decision recommendations
 -  Governance monitoring
 -  Audit and reporting
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3. AI Risk & Decision Lifecycle (High Level)

1. **Risk Assessment (AI-TRUST-001)**
 2. Deterministic scoring using predefined factors
 3. **Behavioral Pattern Detection (AI-TRUST-002)**
 4. Time-based analysis of seller behavior
 5. **Decision Recommendation (AI-TRUST-003)**
 6. Synthesizes risk signals into advisory recommendations
 7. **Human Review & Action**
 8. Humans accept, modify, or reject AI recommendations
 9. **Audit & Oversight (AI-OPS-004/005/006)**
 10. Monitoring, feedback, and controlled evolution
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4. Explainability & Auditability

For every AI evaluation, the system records: - Input data sources used - Applied rules and weights - Intermediate risk calculations - Final recommendation - Confidence score - Human-readable explanation

All records are: - Timestamped - Versioned - Actor-attributed - Retained for up to 7 years

5. Drift Detection & Model Stability

The platform continuously monitors: - Risk score distributions - Escalation frequency changes - Recommendation mix shifts - Behavioral pattern frequency

Comparisons are performed across: - 7-day vs 30-day windows - 30-day vs 90-day windows

Detected drift is: - Classified by severity - Explained with statistical context - Logged for regulatory review

No automatic corrective action is taken.

6. Accuracy & Error Oversight

Post-outcome analysis is used to classify AI decisions into: - Correct Positive - Correct Negative - False Positive - False Negative

Accuracy metrics are produced: - Per risk band - Per rule - Per recommendation type

These metrics are observational only and do not alter live behavior.

7. Human-in-the-Loop Governance

Human operators: - Review AI recommendations - Override decisions when necessary - Provide structured feedback - Remain accountable for final outcomes

The system tracks: - Override rates - Alignment between AI and humans - Decision friction and processing time

8. Controlled AI Change Management

Any modification to AI logic requires: - Formal change proposal - Shadow evaluation against historical data - Impact and safety analysis - Multi-role human approval - Versioned deployment readiness

Emergency rollback can be prepared instantly without live experimentation.

9. Data Protection & Privacy

- No personally identifiable information is exposed in AI logs
- Aggregated and anonymized metrics are used for oversight
- Strict access control and role separation are enforced

10. Regulatory Alignment Summary

This system is designed to align with: - Financial services AI governance expectations - Marketplace trust & safety standards - Model risk management principles - Audit and supervisory review requirements

The platform prioritizes **control, transparency, and accountability** over automation.

Conclusion

The AI Trust & Risk platform functions as a **governed decision-support system**, not an autonomous decision-maker.

It enables scalable trust operations while maintaining: - Human control - Regulatory transparency - Long-term system stability

The design ensures regulators can inspect, audit, and understand every aspect of AI-assisted decision-making.